CheckInn - Hotel Management System

OSDL Mini-Project

Student Details:

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Class & Sec: **CSE-B**

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1. Objective

To design and develop a fully functional Hotel Management System as a desktop application using JavaFX, proving core Object-Oriented Programming principles, including encapsulation, inheritance, polymorphism, generics and multi-threading, while providing a clean, intuitive interface for managing hotel rooms, customers, bookings and revenue analytics with persistent data storage.

2. Technologies used

Technology	Purpose
Java 23	Core programming language
JavaFX 25	Desktop GUI framework
Maven 3.9	Build tool and dependency framework

Java Serialization	Persistent file-based data storage
Java Collections Framework	In-memory data management
CSS3	UI theming
Java Multithreading	Async booking confirmation
Charts.js	Revenue & occupancy visualization

3. Features Implemented

a. Room Management:
   i. Add rooms with room number, type, and price per night
   ii. View all rooms in a styled Table-View
   iii. Filter and display only available rooms
   iv. Color-coded availability status
   v. Save and load room data via room tab

b. Customer Management
   i. Add customers with id, name and contact number
   ii. View all customers with their given room
   iii. Remove customers with confirmation dialog
   iv. Prevents removal of currently checked-in customers

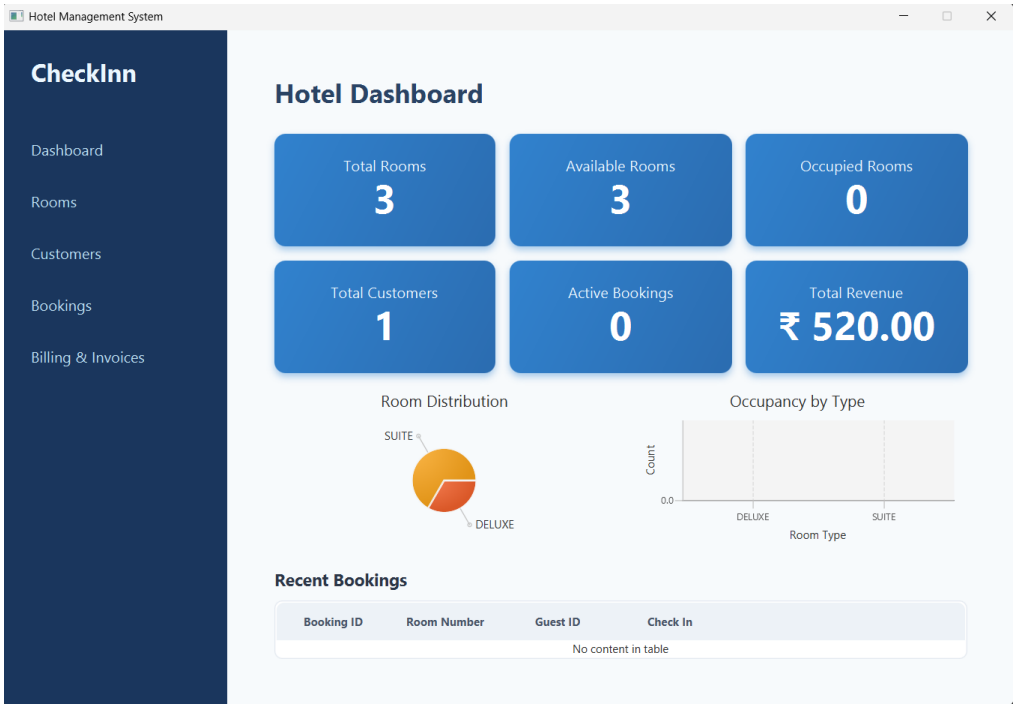
c. Booking & Checkout
   i. Book a room by assigning a customer to an available room
   ii. Select check-in and check-out dates via date-picker
   iii. Automatic total cost calculation based on nights stayed
   iv. Double booking prevention with error alerts
   v. Async booking confirmation with background threads: 1.5s processing simulation
   vi. One-click room checkout with availability auto-update

d. Billing Analytics Dashboard
   i. Summary cards – total revenue, this month, total booking, avg per booking

    ii. Revenue breakdown per room type, with progress bars and %age share
    iii. Monthly revenue bar chart across all 12 months
    iv. List of recent guests and revenue generated
    v. Full bookings revenue table with grand total footer
e. Main Dashboard
    i. Live stat cards: total rooms, available, occupied, customers, bookings, revenue
    ii. Pie chart - room distribution by type
    iii. Bar chart – occupancy breakdown by room type
    iv. Recent bookings list
f. Persistence
    i. Auto-save all data in a “hotel_data.dat” file
    ii. Auto-load all bookings by clicking a button
    iii. Stored in a single file
g. OOP Concepts Demonstrated
    i. Encapsulation – private fields with getters/setters in all model classes
    ii. Inheritance and Polymorphism – Standard, Deluxe and Suite rooms, with overridden tariff calculation
    iii. Enum – room type with base price constructor
    iv. Generics – Pair<T, U> linking room number to guest name in bookings
    v. Multithreading – background thread with platform.runLater() for booking confirmation
    vi. Serialization – full save/load of application state to file

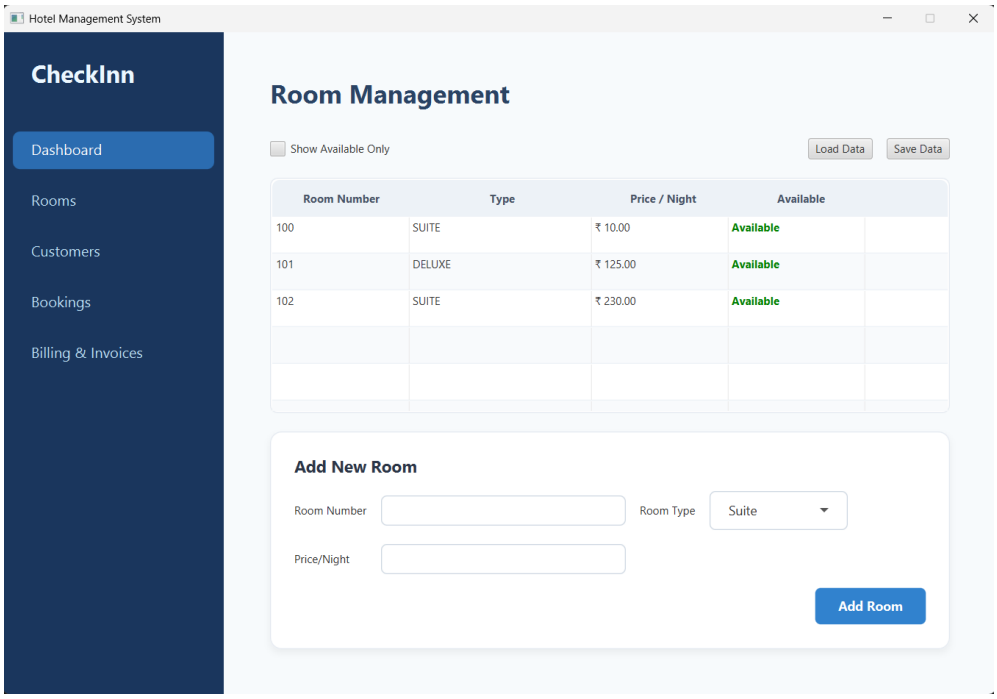
4. Output Screenshots

Main Dashboard:

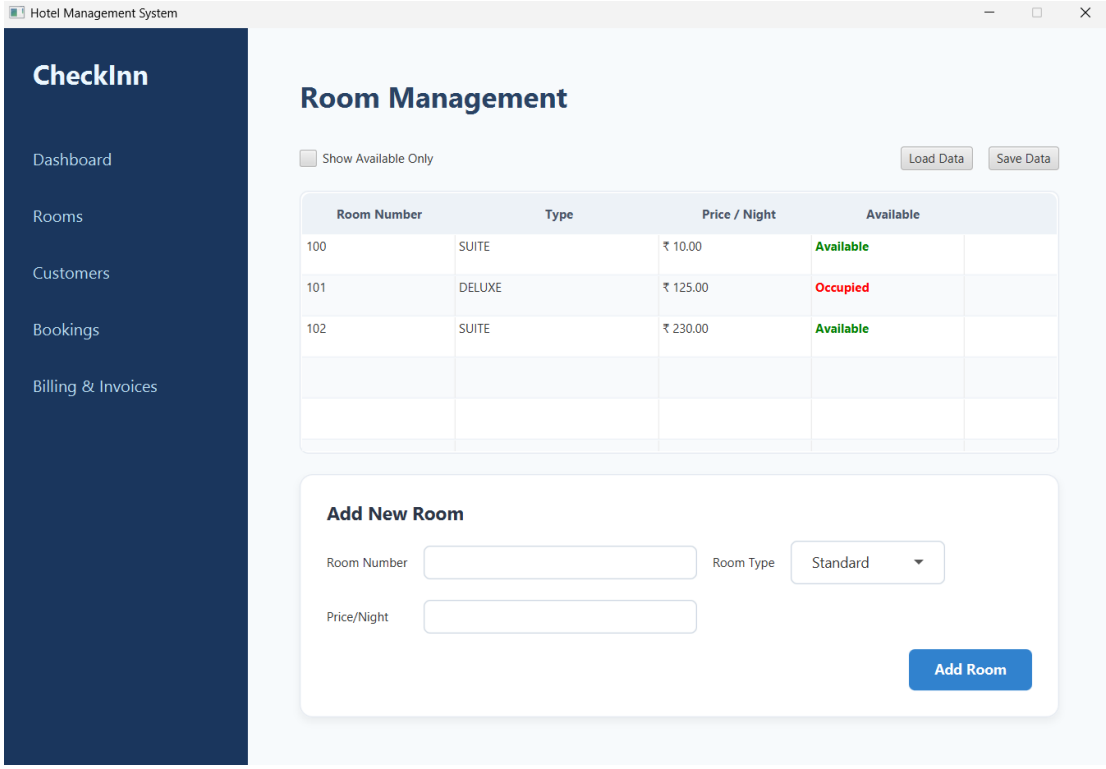


Room Management System:

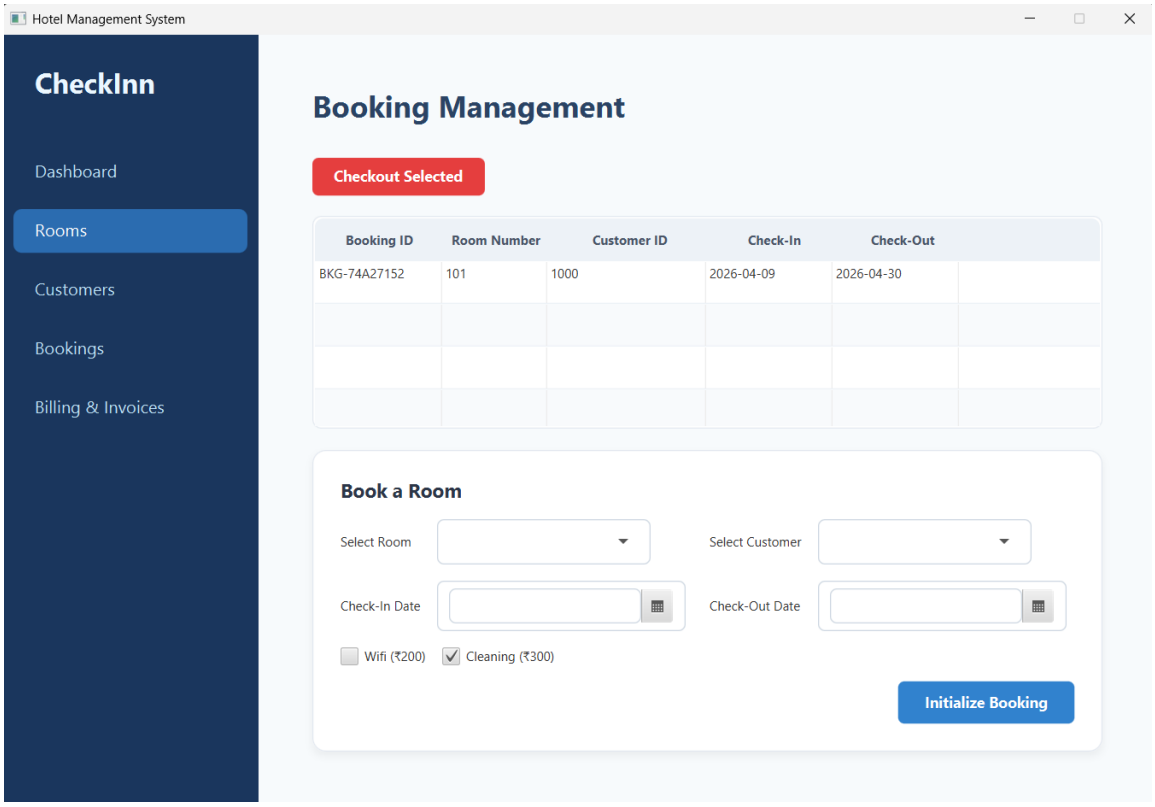
a. Room is available



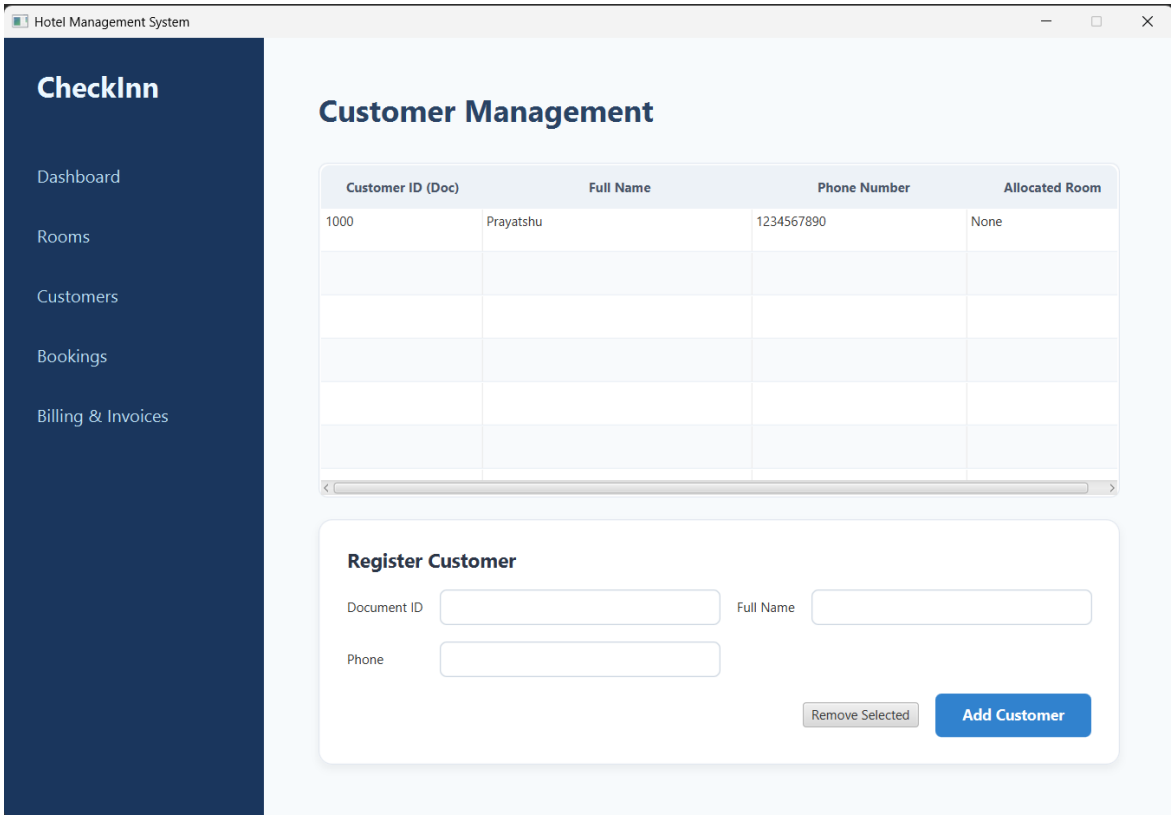
b. One room is occupied



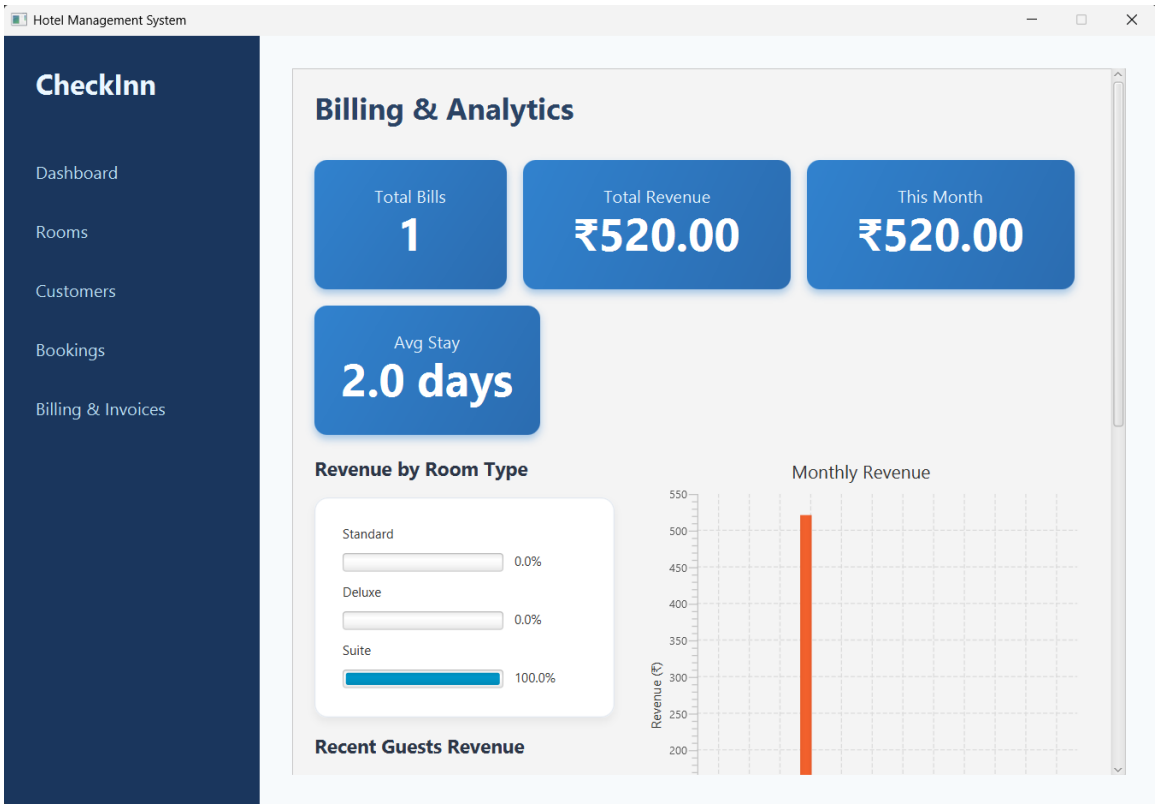
Booking Management System:

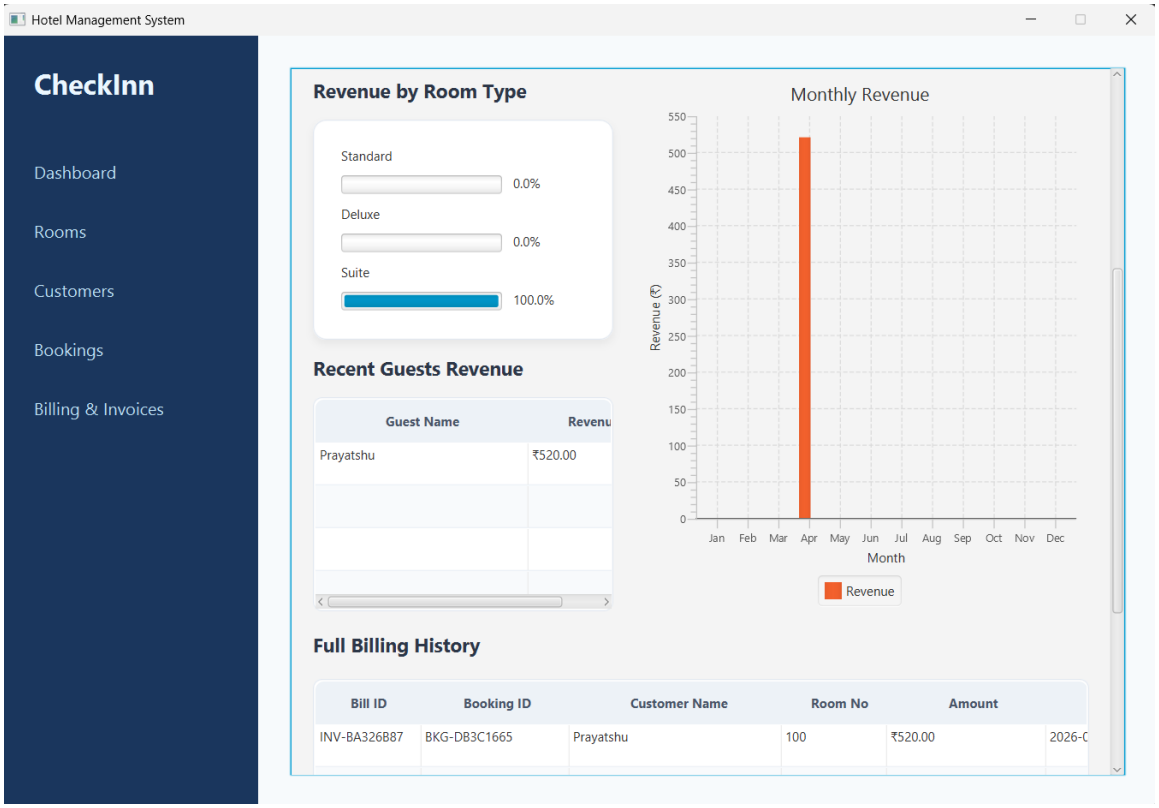


Customer Management System



Billing System





5. Conclusion

This Hotel Management System successfully demonstrates the practical application of Object-Oriented Programming concepts in building a real-world desktop application. By leveraging JavaFX for the GUI and Maven for build management, the project goes beyond the basic lab requirements to deliver a polished, feature-rich system.

The application covers the complete hotel workflow - from room and customer management to bookings, checkout, and revenue analytics - all backed by persistent file storage that ensures data is never lost between sessions. Role-based login adds a layer of access control, making the system closer to a production-ready application.

Key OOP principles including encapsulation, inheritance, polymorphism, generics, multithreading, and serialization are not just theoretically present but actively used throughout the codebase in

meaningful ways. The addition of a Billing Analytics dashboard with charts, a live Dashboard, CSS-based theming with light/dark mode, and a Menu Bar with status bar elevate the project well above standard lab expectations.

Overall, this project serves as a comprehensive demonstration of how core Java and JavaFX concepts can be combined to build a functional, visually appealing, and maintainable desktop application.